Problem 1: Finding Complexity using Counter Method

Started on	Friday, 14 August 2026, 8:32 AM
State	Finished
Completed on	Friday, 14 August 2026, 8:53 AM
Time taken	21 mins 30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increement for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
void function(int n){
    int count = 0;
    int i = 1;
    count++;
    int s=1;
    count++;
    while(s<=n){
        count++;
        i++;
        count++;
        s+=i;
        count++;
    }
    count++;
    printf("%d\n",count);
}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Problem 2: Finding Complexity using Counter method

Started on	Thursday, 17 September 2026, 12:14 PM
State	Finished
Completed on	Thursday, 17 September 2026, 12:16 PM
Time taken	1 min 51 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, count = 0;
    if (scanf("%d", &n) != 1) return 0;

    if (n == 1) {
        count++; // printf("*");
    } else {
        for (int i = 1; i <= n; i++) {
            count++; // Outer loop condition check: i <= n

            for (int j = 1; j <= n; j++) {
                count++; // Inner loop condition check: j <= n
                count++; // printf("*");
                count++; // printf("*");
                count++; // break statement
                break;
            }
        }
        count++; // Outer loop termination check (i = n + 1)
        count++; // Else branch entrance/initialization overhead
    }

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Problem 3: Finding Complexity using Counter Method

Started on	Thursday, 17 September 2026, 12:16 PM
State	Finished
Completed on	Thursday, 17 September 2026, 12:19 PM
Time taken	2 mins 59 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
      {
        printf("%d ", i);
      }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int num, count = 0;
    if (scanf("%d", &num) != 1) return 0;

    count++; // i = 1 initialization
    for (int i = 1; i <= num; ++i) {
        count++; // if (num % i == 0) check
        if (num % i == 0) {
            count++; // printf("%d ", i)
        }
        count++; // ++i increment
    }
    // Loop exit check is accounted for implicitly in initialization/termination step

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Problem 4: Finding Complexity using Counter Method

Started on	Thursday, 17 September 2026, 12:19 PM
State	Finished
Completed on	Thursday, 17 September 2026, 12:21 PM
Time taken	1 min 48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n, count = 0;
    if (scanf("%d", &n) != 1) return 0;

    count++;
    for (int i = n / 2; i < n; i++) {
        count++;
        for (int j = 1; j < n; j = 2 * j) {
            count++;
            for (int k = 1; k < n; k = k * 2) {
                count++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Problem 5: Finding Complexity using counter method

Started on	Thursday, 17 September 2026, 12:21 PM
State	Finished
Completed on	Thursday, 17 September 2026, 12:22 PM
Time taken	46 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n, rev = 0, remainder, count = 0;
    if (scanf("%d", &n) != 1) return 0;

    count++;
    while (n != 0) {
        count++;
        remainder = n % 10;
        count++;
        rev = rev * 10 + remainder;
        count++;
        n /= 10;
        count++;
    }
    count++;
    count++;

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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